For this assignment, I had two primary goals: firstly, to continue the image processing theme that I've been going with in my assignments in the class so far; and secondly, to see if I could investigate what it might be like to use an advanced library for my final project, which has to do with convolutions. Each goal lends itself pretty naturally to a choice of advanced library, NPP for the first and cuBLAS for the second. The combination seemed promising because I wanted to try out NPP to perform some of the image processing operations that I had learned about, and I also wanted to try using GEMM to perform convolutions.

One challenge I faced was making sure that I had all the necessary device buffers allocated with the correct dimensions: there are the buffers for the input image and its transformations, which are simple enough, but then there are some buffers that need some extra care to get their geometry correct, like the buffers for the im2col results and the non-interleaved and interleaved convolution results. I think these challenges were new because I hadn't really written a pipeline that works in device memory for this long.

Another challenge I faced was reshaping some of the intermediate results to fit the libraries' APIs. The toughest case of this that I faced was when combining the results of the x-direction and y-direction Sobel kernel convolution to get a magnitude value for each (x, y) pair: the NPP function that can do this (nppiMagnitude_32fc32f_C1R) expects the inputs to be interleaved like complex numbers, but my memory layout had all the x-direction results before all the y-direction results. Eventually I found out that cudaMemcpy2D can perform copies with stride on both the source and destination, which made it possible to copy the values into a separate buffer in an interleaved fashion. I'd say this was a pretty big triumph.